Industrial Internship Report on

" Obstacle Detection Motor Control"

Prepared by

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time. My project was **"Smart Obstacle Detection and Automatic Motor Control System using Ultrasonic Sensor and Arduino."** The main objective of this project was to detect obstacles in front of a moving system and automatically stop the motor when an object comes within a predefined distance. The system uses an HC-SR04 Ultrasonic Sensor to continuously measure the distance between the sensor and nearby objects. The Arduino processes the sensor data and controls the motor accordingly. This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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This internship was a six-week learning journey that provided me with practical exposure to Embedded Systems and Internet of Things (IoT) technologies. During this period, I worked on the project titled **"Smart Obstacle Detection and Automatic Motor Control System using Ultrasonic Sensor and Arduino."** The internship enabled me to apply theoretical concepts learned during my academic studies to a real-world engineering problem and gain valuable hands-on experience.

Internships play an important role in career development as they bridge the gap between academic knowledge and industry requirements. They provide students with practical experience, improve technical skills, enhance problem-solving abilities, and prepare them for professional careers. Through this internship, I was able to understand how industrial projects are planned, developed, tested, and documented.

The main objective of my project was to develop a system capable of detecting obstacles using an ultrasonic sensor and automatically controlling a motor based on the detected distance. The system continuously monitors the surrounding environment and stops the motor whenever an obstacle is detected within a predefined range. This helps improve safety and demonstrates the application of embedded systems in automation and robotics.

This opportunity was provided through the collaboration of **upskill Campus (USC)**, **The IoT Academy**, and **UniConverge Technologies Pvt. Ltd. (UCT)**. The internship program offered valuable guidance, learning resources, project-based training, and industry-oriented exposure that helped me understand current technological trends and practical applications.

The program was systematically planned over six weeks. Initially, I studied the fundamentals of embedded systems, sensors, and Arduino programming. In the following weeks, I designed the system architecture, implemented the project, tested the hardware and software components, and analyzed the performance of the system. Finally, I prepared the project documentation and report summarizing the complete work.

Throughout this internship, I learned Arduino programming, sensor interfacing, motor control, embedded system design, circuit implementation, debugging techniques, and project documentation. I also improved my analytical thinking, problem-solving skills, teamwork, communication abilities, and technical confidence. Overall, this internship was an enriching and rewarding experience that contributed significantly to my professional growth.

I would like to express my sincere gratitude to **upskill Campus (USC)**, **The IoT Academy**, and **UniConverge Technologies Pvt. Ltd. (UCT)** for providing this valuable opportunity. I would also like to thank my mentors, trainers, faculty members, and everyone who directly or indirectly supported me throughout the internship. Their guidance and encouragement played an important role in the successful completion of my project.

3 Introduction

3.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

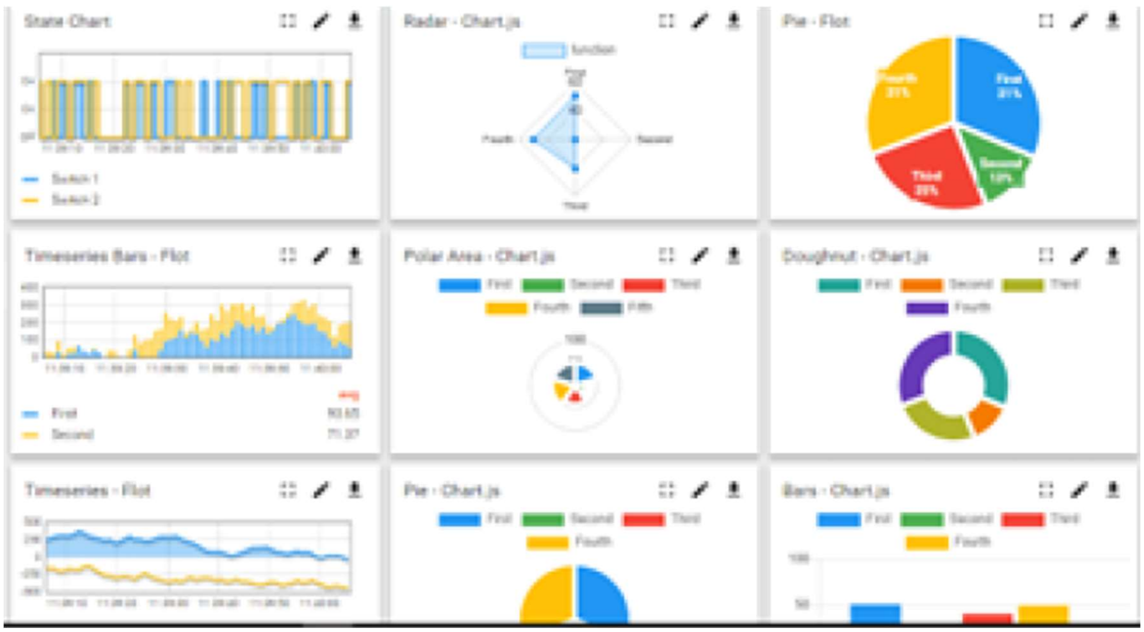
i. UCT IoT Platform ( )

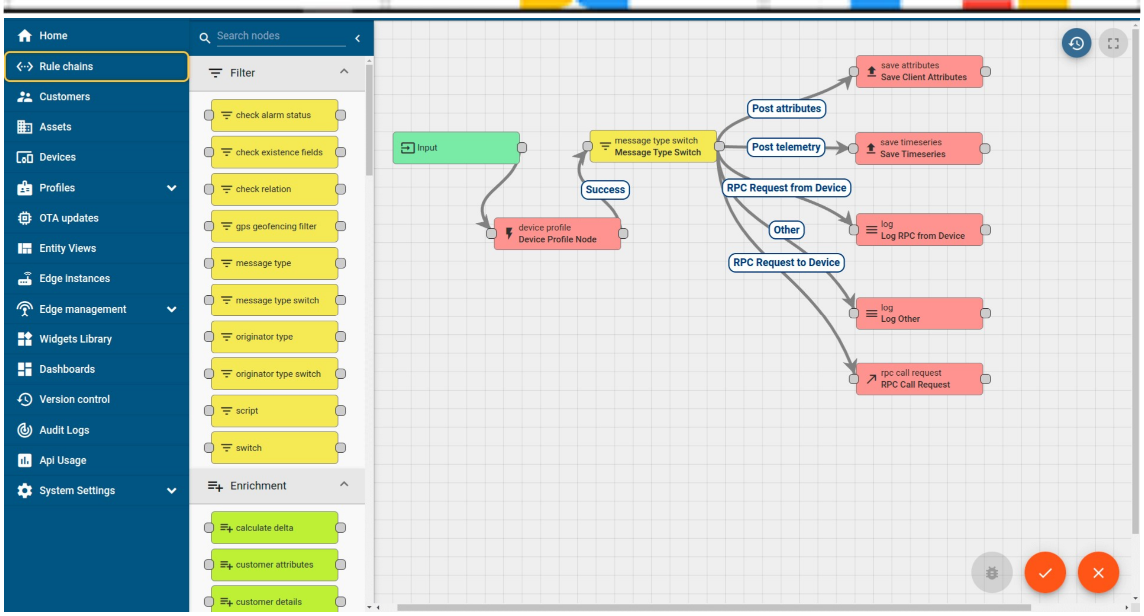
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
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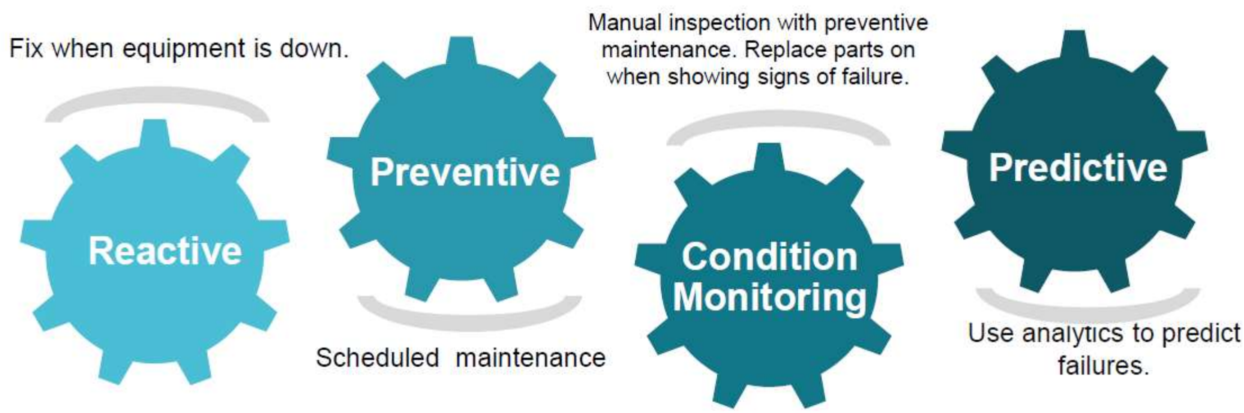
iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

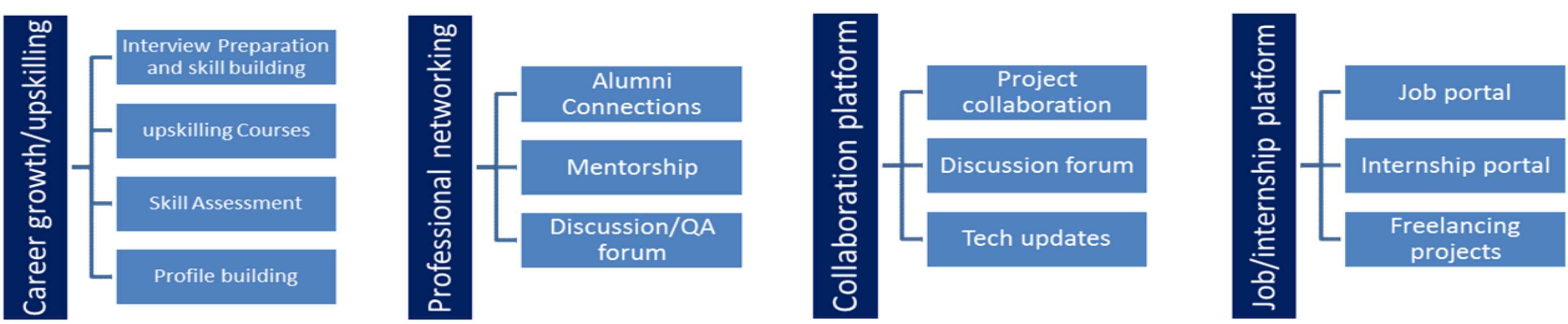
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



3.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



3.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

3.4 Objectives of this Internship program

The objective for this internship program was to

☛ get practical experience of working in the industry.

☛ to solve real world problems.

☛ to have improved job prospects.

☛ to have Improved understanding of our field and its applications.

☛ to have Personal growth like better communication and problem solving.

3.5 Reference

[1] Arduino Official Documentation – https://www.arduino.cc

[2] HC-SR04 Ultrasonic Sensor Datasheet

[3] Arduino IDE Documentation

3.6 Glossary

Terms	**Acronym**
Arduino Integrated Development Environment	IDE
Ultrasonic Sensor	US
Direct Current Motor	DC Motor
Trigger Pin	TRIG
Echo Pin	ECHO

Internet of Things	IoT

4 Problem Statement

In the assigned problem statement

The assigned problem statement was to develop an embedded system capable of detecting obstacles and automatically controlling a motor based on the distance measured by an ultrasonic sensor. In many automation, robotics, and industrial applications, motors continue operating even when obstacles are present, which can lead to collisions, equipment damage, or safety issues.

To address this problem, a Smart Obstacle Detection and Automatic Motor Control System was designed using an HC-SR04 Ultrasonic Sensor and Arduino. The system continuously measures the distance between the sensor and nearby objects. When an obstacle is detected within a predefined distance (20 cm), the motor is automatically turned OFF. When the obstacle is removed and the path becomes clear, the motor resumes operation.

The proposed solution improves safety, reduces the risk of accidents, and demonstrates the practical application of embedded systems in automation and intelligent control systems.

5 Existing and Proposed solution

Provide summary of existing solutions provided by others, what are their limitations?

Existing obstacle detection systems are commonly used in industrial automation, robotics, and smart vehicles. Many traditional systems rely on manual monitoring or simple mechanical switches to detect obstacles. These solutions often have limitations such as delayed response, limited detection range, higher maintenance requirements, and reduced reliability in dynamic environments.

The proposed solution is a **Smart Obstacle Detection and Automatic Motor Control System** using an Arduino and HC-SR04 Ultrasonic Sensor. The system continuously measures the distance between the sensor and nearby objects. When an obstacle is detected within a predefined range (20 cm), the motor automatically stops. Once the obstacle is removed, the motor resumes operation.

The main value addition of the proposed system is its automatic operation, low cost, real-time obstacle detection, improved safety, and ease of implementation. The system can be further extended for use in smart vehicles, robotic systems, industrial conveyor belts, and IoT-based automation applications.

5.1 Code submission (Github link)

upskillcampus/ObstacleDetectionMotorControlProject.ino at main · Dhruv2436/upskillcampus

5.2 Report submission (Github link) : first make placeholder, copy the link.

5.3 upskillcampus/ObstacleDetectionMotorControl_Dhruv_USC_UCT.pdf at main · Dhruv2436/upskillcampus

6 Proposed Design/ Model

7 Proposed Design / Model

The proposed system consists of an Arduino Uno, an HC-SR04 Ultrasonic Sensor, and a DC Motor. The ultrasonic sensor continuously measures the distance between the system and nearby objects. The Arduino processes the sensor data and compares the measured distance with a predefined threshold value.

If the detected distance is less than 20 cm, the Arduino automatically stops the motor to avoid collision and ensure safety. When the obstacle is removed and the distance becomes greater than 20 cm, the motor resumes operation. The system operates in real time and continuously monitors the surrounding environment.

8 Design Flow

1. Start the system.
2. Initialize Arduino, Ultrasonic Sensor, and Motor.
3. Measure the distance using the HC-SR04 sensor.
4. Compare the measured distance with the threshold value (20 cm).
5. If distance < 20 cm, turn the motor OFF.
6. If distance ≥ 20 cm, turn the motor ON.
7. Repeat the process continuously for real-time monitoring.

9 Final Outcome

The developed system successfully detects obstacles and automatically controls the motor based on the measured distance. This improves safety, reduces the risk of collisions, and demonstrates the practical implementation of embedded systems in automation and robotics.

10 Performance Test

The performance of the Smart Obstacle Detection and Automatic Motor Control System was evaluated by testing the accuracy and response of the ultrasonic sensor and motor control mechanism. The main constraints considered during testing were distance measurement accuracy, response time, power consumption, and reliability.

The HC-SR04 ultrasonic sensor provides accurate distance measurements within its operating range. The Arduino processes the sensor data quickly and controls the motor with minimal delay. The system was designed to continuously monitor obstacles and respond immediately when an object is detected within 20 cm.

10.1 Test Plan/ Test Cases

Test Case ID	Test Scenario	Input Condition	Expected Result	Status
TC-01	Normal Operation	No obstacle within 20 cm	Motor ON	Passed
TC-02	Obstacle Detection	Obstacle at 15 cm	Motor OFF	Passed
TC-03	Safe Distance Check	Obstacle at 25 cm	Motor ON	Passed
TC-04	Dynamic Object Movement	Object moves continuously	Motor responds in real time	Passed
TC-05	Reliability Test	Repeated execution of system	Stable and consistent operation	Passed

10.2 Test Procedure

1. Power ON the Arduino and sensor circuit.
2. Place an object at different distances from the ultrasonic sensor.
3. Observe the measured distance on the Serial Monitor.
4. Verify motor operation based on the distance threshold.
5. Repeat the test multiple times to ensure consistent performance.

6. Record the observations and compare them with expected results.

10.3 Performance Outcome

The developed system successfully detected obstacles and controlled the motor according to the predefined distance threshold of 20 cm. The response time was fast enough for real-time operation, and the sensor provided reliable distance measurements. The system operated consistently during repeated testing without significant errors.

11 My learnings

This internship provided me with valuable practical knowledge of Embedded Systems and IoT concepts. During the development of the Smart Obstacle Detection and Automatic Motor Control System, I learned how to interface an ultrasonic sensor with an Arduino, process real-time sensor data, and control hardware components based on programmed conditions.

I gained hands-on experience in Arduino programming, circuit design, hardware integration, troubleshooting, and system testing. I also improved my problem-solving, analytical thinking, and debugging skills while working on the project. Additionally, I learned the importance of project planning, documentation, testing procedures, and performance evaluation in engineering projects.

This internship helped me bridge the gap between theoretical concepts and practical implementation. The knowledge and skills gained through this experience will contribute significantly to my career growth in the fields of Embedded Systems, IoT, Automation, Robotics, and Software Development. Overall, the internship enhanced my technical confidence and prepared me for future industry challenges.

12 Future work scope

The current system successfully detects obstacles and controls the motor automatically based on the measured distance. However, several improvements can be made in the future to enhance its functionality and industrial applicability.

Future enhancements may include integrating IoT technology to enable remote monitoring and control through a mobile application or web dashboard. Additional sensors such as infrared sensors, cameras, or proximity sensors can be incorporated to improve detection accuracy and reliability. The system can also be extended to support autonomous robot navigation and smart vehicle applications.

Furthermore, features such as buzzer alerts, LED indicators, real-time data logging, and cloud-based monitoring can be added to improve user interaction and system performance. These enhancements would make the project more suitable for advanced automation, robotics, and industrial safety applications.